Summer 2026: Progress and Results

QUANTT Credit Trading

Period 19 July to 16 August 2026 · issued 16 August 2026

github.com/quanttqueensu/CreditTrading

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1. Summary

We set out to build a credit strategy that performs independently of market direction. We tested thirteen mechanisms. Twelve failed. The survivor is deployed on a $500,000 Interactive Brokers paper account and places its own orders on a schedule.

We also built a free daily feed of 11,423 bond prices, established that our own cost model had overcharged every strategy by a factor of 3.7, identified sixteen defects in our own work, and established on the first live trading day that execution cost is a larger constraint than signal quality.

The ratio of thirteen tested to one deployed is the most representative figure for the summer.

2. What we built

2.1 The strategy

Credit closed-end fund discount reversion. A closed-end fund trades on an exchange with a permanently fixed share count, so no mechanism pulls its price back to the value of its holdings. Credit closed-end funds trade at a mean discount of 3.16% with a standard deviation of 5.95%, approximately 150 times the equivalent ETF gap.

The strategy z-scores each fund's discount against its own 252-day history, holds the cheapest long and the richest short in equal dollar amounts, and targets 6% annualised volatility.

2.2 Infrastructure

We built a backtest engine with a lookahead guard and purged walk-forward testing; a deployment framework in which every strategy implements the same four methods; and 1,077 lines of Interactive Brokers integration including a gate that reconciles our records against the broker's before any order transmits. Each strategy maintains a ledger recording positions, orders, modelled trades, real fills and slippage. Five scheduled jobs run the book unattended on weekdays, behind seven preflight checks, against a cost model covering 45 tickers calibrated to measured bid-ask spreads.

2.3 Data

DATASET	SIZE	NOTE
Daily bond price panel	11,423 bonds/day	Free, from fund holdings files. No archive exists, so it accumulates forward only

DATASET	SIZE	NOTE
SEC N-PORT holdings	136,268 rows from 2019	95.3% CUSIP overlap against a live holdings file
CEF prices and NAVs	265,615 and 232,799 rows from 1986	44 funds
Fallen-angel events	16,388 migrations, 2003–2025	With index removal dates
FINRA short volume	26,116 rows	Daily, free
Tradable universe	10 to 56 instruments	

3. Results

3.1 The thirteen mechanisms

MECHANISM	VERDICT	CAUSE
credit_rv	Killed	Sealed holdout Sharpe −1.44; edge negative before costs
E1, HYG vs JNK	Killed	Negative out of sample; opportunity fell from 188bp to 3.8bp
S1, cross-issuer disagreement	Killed	Input carries no information: median disagreement 0.09bp, as all issuers price from one vendor
S3, fallen angels via ETFs	Killed	A credit-quality premium in disguise. Unconditional holding scored 0.37; signal-conditioned scored 0.03 to 0.24
Single-fund premium/discount	Killed	Predicts the NAV converging, not the price. Trading it opposes price discovery
leadlag	Killed	Measured at a non-transactable price. At tradable prices t fell from 5.20 to −0.50, and the control group scored higher
nport-trace-nav	Infeasible	Only 17 to 30% of holdings trade daily, and those are selected for news
dealer-constraint	Killed	No credit name significant at any horizon; Treasury control showed equal magnitude

MECHANISM	VERDICT	CAUSE
pair-reversion	Killed on arithmetic	Gross Sharpe +1.03, but costs do not diversify while volatility does, giving net −0.16. Volatility target required 32.8× leverage against a 2× limit
short-pressure	Killed	Rates control group scored higher than credit
raw-flow-z	Killed	Precise zero across 28,025 observations, all t below 0.8
Distribution events	Killed	Cuts move the discount against the thesis and show no dose-response
CEF discount reversion	**Deployed**	Passed the full battery below

Two mechanisms remain on watch rather than killed. Forced-flow price pressure is real and large, at −424bp with $t = -17.5$ and 82 to 85% reversal, but the only available expression dilutes the signal fivefold. The premium/discount band form retained its edge per opportunity; only the opportunity count collapsed.

3.2 Validation of the deployed strategy

TEST	RESULT	VERDICT
Point-in-time universe, no survivorship or liquidity hindsight	Gross 1.26, net 0.82, volatility 6.00%, max drawdown −12.0%	Pass, and better than the biased construction
Purged walk-forward, 10 blocks, 5-day embargo	9 of 9 positive, median 1.12, worst 0.01	Pass
Block bootstrap, 5,000 draws	5th/95th percentile 0.52/1.11, P(SR ≤ 0) = 0.000%	Pass
Deflated Sharpe, adjusted for 10 specifications	0.956	Pass
Factor exposure	Alpha t 3.11, R^2 0.005, 5 of 5 factor limits	Pass

An R^2 of 0.005 means 99.5% of the return is unexplained by high yield, investment grade, rates, equity or volatility.

3.3 Two attempts to invalidate the result

The live book is 66% net short municipal funds and 57% net long taxable funds, so the natural hypothesis was a disguised sector position, which is what one earlier candidate proved to be.

CONTROL SET	ALPHA P.A.	T	R^2
Base factor set	+8.21%	+5.67	0.0037
Plus municipal ETFs	+8.88%	+5.69	0.0117
Plus municipal ETFs and duration spread	+8.83%	+5.67	0.0125
All five CEF group factors	+7.63%	+5.90	0.0057

The final row is decisive, since municipal CEF discounts do not track municipal ETFs. Under it every group beta is at or below 0.025 and the alpha is unchanged.

3.4 Three defects in the deployed configuration

An end-to-end audit on 31 July identified three problems. They alter the expected result and are therefore reported here.

The specification, the code and the optimum disagreed on rebalance frequency. The spec declared five days, only an unrelated file read that key, so the live code traded every session, and the measured optimum was two days. Net Sharpe runs 0.62 at one day, 0.73 at two, 0.51 at five and 0.20 at twenty-one. We changed it to two.

The backtest assumed an unobtainable entry price, entering at day *t*'s close using day *t*'s NAV, which publishes after that close. A market-on-close order fills at *t+1*'s close. The correction costs 38% of the Sharpe, taking 0.82 to 0.51.

No sealed holdout was taken for this strategy. The killed credit_rv strategy received 141 trials and a sealed holdout before being terminated; this one received ten specifications and $500,000 within two hours. That was a governance failure, and it occurred on the only strategy that received capital.

The honest expected net Sharpe of the deployed configuration is therefore 0.51, not 0.82. Correcting the holding period raises it to approximately 0.73.

3.5 The holdout we opened and failed

On 31 July we opened a sealed holdout on a proposed change, a 63-day z-window in place of 252, against a pass mark of +0.40 recorded in advance. It scored −0.298 across 646 days of 2024 to 2026 data, and we reverted the change.

The signal itself generalised: gross Sharpe rose from 1.23 in sample to 1.75 out of sample, the best gross figure the project has produced. Turnover was the failure, at 102.6 times per year against 45.3 in sample, after which costs consumed 117% of gross.

The finding generalises to every configuration decision we make. Selecting a configuration on net Sharpe is implicitly selecting it on a turnover estimate, and turnover is substantially less stable out of sample than the signal. This failure mode operates with the alpha fully intact, and is most severe on fast configurations where turnover is largest.

4. Live trading record

The strategy funded on 30 July and traded on 31 July, which remains the only trading day.

BOOK	CAPITAL	GROSS	P&L	RETURN
cef_discount	$500,000	$749,458	–$7,350	–1.47%
phase0_null_trader	$640,000	$640,000	–$281	–0.04%
Five benchmark books	$20,000 each	$100,000	–$253 total	

Attribution summed to −$7,884 against the broker's reported −$7,883.09, so every position is accounted for.

4.1 Execution finding

Priced at the decision prices the strategy traded on, the book was up $50. Priced at the fills we received, it was down $7,350. The difference is entirely slippage.

The strategy must decide in the evening, because its signal requires the NAV, which publishes after the close. Plain market orders therefore rested overnight and filled at 07:27 ET, two hours before the exchange opened, in funds with negligible pre-market depth.

TRADED	SLIPPAGE	REALISED	MODELLED	RATIO
$682,351	$6,405	0.94%	~0.10%	9.4×

Slippage ran against us on every buy and every sell, the signature of crossing a wide spread rather than noise. At 24 rebalances per year this is 22.6% annually against a strategy earning 4.85%.

We excluded the possibility of erroneous prices by reconciling all 17 funds against the broker's daily bars over ten days; median disagreement was 0.000%. The cause is purely execution.

We switched to market-on-close orders, which execute in the closing auction, the deepest liquidity of the day and the point the backtest assumed. Routing was verified live and the scheduled job transmitted 16 market-on-close orders.

Whether this is sufficient is the most important open question in the project, ahead of returns. We cannot slow the strategy to escape the cost, as net Sharpe falls sharply with holding period.

4.2 Current status

The system has not traded since 1 August. Interactive Brokers TWS has not been running, and every session since ends with the blocker “nothing listening on 127.0.0.1:7497”. TWS restarts daily and requires an interactive login.

We therefore hold one day of live evidence, not sixty. The 60-session review required by the kill rule remains ahead of us, and restoring the broker connection is our first priority.

The remainder of the system operated correctly throughout the outage. Data collection, reporting and the watchdog have run every weekday, which is the four-phase session design performing as designed: a book that cannot trade still records.

5. Evidence

All claims above are verifiable against files in the repository.

5.1 Broker executions

`ops/books/cef_live/_ibkr_shadow/cef_discount/broker_fills.csv` holds 257 real executions from the paper account, each with the broker’s execution id, commission and timestamp:

```
2026-07-31T21:20:29.234516+00:00,2026-07-31,BIT,BUY,720.0,12.55,ibkr_paper,
execId=00012ec5.6a6cade0.01.01 orderRef='' commission=3.60216
time=2026-07-31 11:27:57+00:00
```

The 11:27 UTC timestamp is the 07:27 ET pre-market fill described in section 4.1, so the evidence of the defect and the record of the trade are the same file. `ops/books/phase0_live/_ibkr_shadow/null_trader/broker_fills.csv` holds a further 45 executions for the control experiment.

5.2 Ledger trail

Each book maintains `nav.csv`, `positions.csv`, `orders.csv`, `trades.csv`, `slippage.csv` and `manifest.json`, the last verified on every load. The slippage file records the execution failure by name: AWF at 158.6bp realised against 17.5 modelled, BIT at 261.7 against 13.8, PFN at 254.2 against 18.8.

5.3 Trial ledger

`results/credit_rv/trial_log.csv` holds 156 numbered trials, each timestamped, with Sharpe, return, volatility, drawdown, turnover, cost drag and a written note. Trial 1 is dated 2026-07-28 20:12:03 and reads “first baseline: continuous sizing, 1993+ sample, 87x/yr turnover, rejected on cost". Trial 156 is dated 2026-07-31 01:44:18. This file is our multiple-testing count, it never resets, and it is what makes the deflated Sharpe calculation legitimate.

5.4 Holdout records

`results/cef/HOLDOUT_PREREG.md` states the rules and pass mark and was written before the holdout was opened. `results/cef/HOLDOUT_OPENED.json` records the opening at 2026-07-31 21:47 UTC, the window, the result of −0.298 and the verdict FAIL. `results/credit_rv/HOLDOUT_OPENED.json` records the holdout that killed the earlier strategy at −1.435, $t = -2.29$.

A holdout that fails and is recorded as failing is stronger evidence of process integrity than one that passes.

5.5 Operational record

`ops/schedule/logs/` holds 42 session logs from 20 July to 15 August, one per job per day across five job types. `ops/heartbeat.json` records per-job liveness; at issue it shows collection succeeding 15 August 04:28, the weekly report at 09:02, and both trading jobs correctly reporting themselves blocked by the absent broker connection.

5.6 Pre-commitment documents

`CREDIT_RV_PREREG.md` and `E1_PREREG.md` are pre-registrations for two strategies subsequently killed, stating success criteria before the tests were run. Files in `ops/specs/` are frozen specifications, each carrying an evidence block and a kill rule committed in advance. Every parameter subsequently changed carries a note field recording the measurement behind the change.

6. Parallel research programme

Research ran as concurrent independent investigations rather than a single sequential thread, and each left its own artifacts.

The clearest instance is the second research night, when four investigations ran in parallel alongside the critical path. Three returned useful results, two of which corrected our own assumptions.

The government filings investigation produced 136,268 rows of holdings history and corrected us on two points: the public filings contain one snapshot per quarter rather than three monthly ones, and the valuation-difficulty field we intended to use is degenerate, with 135,850 of 135,870 corporate bonds carrying an identical classification.

The universe investigation expanded us from 10 tradable instruments to 56, and separately identified that the final bar in our price panel was a partial intraday capture, which would have introduced a false return on the most recent day.

The positioning investigation delivered 26,116 rows of short-selling data and a useful negative: our financing file contains no real borrowing-cost data, so any strategy built on borrowing costs was never feasible from our holdings. Full account in `HOW_WE_GOT_HERE.md` section 3.2.

The structure is visible in the repository. `results/` holds one directory per investigation family — `s1`, `s3`, `s4`, `disp`, `leadlag`, `ou`, `positioning`, `universe`, `e1`, `credit_rv`, `cef`, `bench` — mirrored in `scripts/`. These were produced concurrently.

Overnight sessions left their own record. `results/journal/2026-07-30_overnight.md` is a checkpoint journal written, in its own words, so that "the morning reader can reconstruct the night without reading logs". It records each gate with figures attached: 15 funds, 29,698 holdings, 11,423 priced bonds, two incorrect fund ids caught, NAV rebuild matching within 0.07% on 10 of 12 funds, and two accounting defects found and fixed. `results/OVERNIGHT_REPORT_2026-07-30.md` is the full write-up.

Separate lines of work produced their own analysis documents with reproduction scripts: `results/AUDIT_2026-07-31.md`, `results/ACADEMIC_REPORT_2026-07-31.md`, `results/cef/research/NICHE_EDGE_RESEARCH.md`, `results/cef/ESTIMATOR_NOTE.md`, `results/cef/DIST_CUT_NOTE.md`, `results/credit_rv/FINDINGS.md` and `results/e1/E1_RESEARCH_NOTE.md`.

Handoff boundaries are marked in the code. `ops/schedule/install.sh` records that the agent which built it ran only the default render mode, and that installing and enabling the scheduled jobs is left to a human. `RESEARCH_STATE.md` credits agent staging for the N-PORT and breadth datasets and maintains a separate trial counter per data source.

7. Defects identified in our own work

#	DEFECT	DETECTION
1	Compared a full bid-ask spread to a half spread, halving the reported cost ratio	Re-derived the comparison
2	Two fund ids pointed at wrong funds, one an equity fund	Printed the column set on first pull
3	Event study showed a 414% recovery	Impossible on its face
4	Survivorship: bonds that ceased trading dropped out	Re-ran under five sample definitions

#	DEFECT	DETECTION
5	Counted trading days rather than calendar days, so “60 days” meant a year for illiquid bonds	Same re-examination
6	Interest on idle cash counted as skill	Costs came out negative
7	Charged financing to self-funding market-neutral books	Random control scored −0.77 where zero is required
8	Measured returns at a non-existent price	Re-tested at tradable prices; effect vanished
9	Used a varying hedge ratio on both sides, creating false momentum	Control group scored highest
10	Assumed dislocation implies opportunity	Measured by regime rather than assumed
11	Assumed group-neutral matching would help	Tested one change at a time
12	Selected the fund list using current data	Rebuilt point-in-time
13	Broker never read its own configuration; every live run would have failed	Attempted a real connection
14	Position filter ran after neutralisation, leaving the book lopsided	Checked the sum of the first live order set
15	Scheduler invoked a non-existent command	Tested the scheduled path, not the strategy
16	Reported the control experiment as trading when it had only dry-run	Read the log rather than trusting our summary

Six of these — 3, 6, 7, 8, 9 and 13 — were detected because a control or sanity check produced an impossible value. Negative controls do not find the edge; they identify the occasions on which we were about to mislead ourselves.

8. Transferable findings

Costs do not diversify while volatility does. Adding strategies reduces volatility, but costs are deducted in full from each, so cost drag measured in Sharpe grows with the number of strategies. This killed our strongest combination result, which showed a gross Sharpe of +1.03 and a net of −0.16.

Selecting on net Sharpe means selecting on a turnover estimate, and turnover is the unstable component. Our sealed holdout failed on exactly this while the signal improved out of sample.

Testing must cover the deployed path, not the logic alone. Three defects — the broker port, the scheduler flag and the missing strategy class — were invisible to research testing and would have surfaced only as live failures.

Data availability must be confirmed before design. We built an entire research session around holdings history before establishing that no archive exists.

Favourable results warrant the most scepticism. Every false discovery we made appeared excellent initially, with t-statistics of 25 and recoveries of 414%.

A strategy is valued by its contribution to the book, not its standalone score. A mediocre strategy uncorrelated with existing holdings is worth more than a strong one that moves with them.

9. Position entering the school year

The signal is established and survived every control we could construct. The infrastructure schedules, checks, halts and reports itself, including reporting its own failures.

Six items are open, in priority order.

1. Establish whether market-on-close brings slippage within tolerance. We hold no sessions of evidence on this and everything depends on it.
2. Restore a reliable broker connection so sessions trade. One day of live data is not a track record.
3. Reach 60 sessions and execute the pre-committed review.
4. Build a defence against distribution cuts, which permanently re-rate a discount wider and are the standard loss mode for this strategy.
5. Resize the books, currently 158% committed against account equity.
6. Pre-register the January CEF basis trade. It shows +11.71bp per day in January at $t = 3.99$, was positive in 20 of 23 Januaries, and correlates −0.001 with the deployed strategy. Combining 0.82 and 0.73 at zero correlation gives 1.10, a larger gain than any realistic improvement to the main strategy, and it is five months away.

We hold one established signal, one functioning system, one day of live evidence, and one unresolved question over whether the strategy is affordable to trade.